Generalized Matrix Nearness Problem

Projection

$$\min \|A - X\| \quad \text{s.t. } X \in S \qquad X^{\#} = \mathrm{Proj}_S(A)$$

Low Rank

$$\min \|A - X\| \quad \text{s.t. } \mathrm{rank}(X) \le r$$

$$X^{\#} = \mathrm{Proj}_{M_r}(A) = \mathrm{tSVD}_r(A)$$

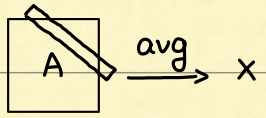
Linear Structure

e.g. Toeplitz

$$\min \|A - X\| \quad \text{s.t. } X \in \mathcal{L}$$

$$\mathcal{L} = \left\{ X = \sum_{i=1}^{s} x_i \underbrace{U_i}_{} \right\} \quad \text{Basis}$$

$$U_1 = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \cdots \quad U_s = \begin{bmatrix} & & & 1 \\ & & 1 & \\ & 1 & & \\ 1 & & & \end{bmatrix}$$



PSD Symmetric

$$\min \|A - X\| \quad \text{s.t. } X \in S^{+}$$

$$X^{\#} = \mathrm{Proj}_{S^+}(A) = \left[\frac{A + A^H}{2}\right]_+ = [\mathrm{Sym}(A)]_+$$

$$X^{\#} = \mathrm{Proj}^{W}_{S^+}(A) = W^{-1/2}[W^{1/2}\,\mathrm{Sym}(A)\,W^{1/2}]_+ W^{-1/2}$$

$$X^{\#} = \mathrm{Proj}^{H}_{S^+}(A) = [\mathrm{Sym}(A \circ H)]_+$$

$\min \ \|A-X\| \quad \text{s.t.} \ X \in S_1 \cap S_2$

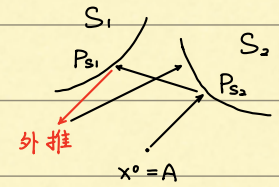
AP (alternative projection)

$X^{k+1} = P_{S_2}(P_{S_1}(X^k))$ 会振荡 / 不一定收敛到 $\operatorname{argmin} \|A-X\|$

Dykstra

$Y^{k+1} = \mathrm{Proj}_{S_1}(X^k - Z^k)$ $\qquad$ $X^{k+1} = \mathrm{Proj}_{S_2}(Y^{k+1} - W^k)$

$Z^{k+1} = Z^k - Y^{k+1} + X^k$ $\qquad$ $W^{k+1} = W^k - X^{k+1} + Y^{k+1}$



S_1, S_2 convex 必收敛

ADMM

$\min \ \|A-X\| \quad \text{s.t.} \ X \in S_1 \cap S_2$

$\min \ \|A-X\|^2 + I_{S_1}(X) + I_{S_2}(Y) \quad \text{s.t.} \ X = Y$

$AL = \|A-X\|^2 + I_{S_1}(X) + I_{S_2}(Y)$

$\quad + \langle Z, X-Y \rangle + \frac{1}{2}\rho \|X-Y\|^2$

$= \|A-X\|^2 + I_{S_1}(X) + I_{S_2}(Y) + \frac{1}{2}\rho \|X-Y+\frac{1}{\rho}Z\|^2 - \frac{1}{2\rho}\|Z\|^2$

$X^{k+1} = \operatorname{argmin} \|A-X\|^2 + \frac{1}{2}\rho \|X-Y+\frac{1}{\rho}Z\|^2 \quad \text{s.t.} \ X \in S_1$

$= \mathrm{Proj}_{S_1}\left(\frac{2}{2+\rho}\left(A + \frac{1}{2}\rho Y^k - \frac{1}{2}Z^k\right)\right)$

$Y^{k+1} = \operatorname{argmin} \|X-Y+\frac{1}{\rho}Z\|^2 \quad \text{s.t.} \ Y \in S_2$

$= \mathrm{Proj}_{S_2}(X^{k+1} + \frac{1}{\rho}Z^k)$

$Z^{k+1} = Z^k + X^{k+1} - Y^{k+1}$

当 $\rho = 1$ 时，且 S_1 为线性空间，ADMM 即 Dykstra (?)

Generalized Nearness

$\min \|A - BXC\|$ 可用 SVD 转化

$\min \|A - BXC\|$ s.t. $\mathrm{rank}(X) \le r$ F&T

$$\underset{m\times p}{B} = \underset{m\times m}{U_B} \underset{m\times p}{\begin{bmatrix} \Sigma_B & 0 \\ 0 & 0 \end{bmatrix}} \underset{p\times p}{V_B^T} \qquad \underset{q\times n}{C} = \underset{q\times q}{U_C} \begin{bmatrix} \Sigma_C & 0 \\ 0 & 0 \end{bmatrix} \underset{n\times n}{V_C^T}$$

$\mathrm{rank}(B) = s$ $\qquad$ $\mathrm{rank}(C) = t$

$$U_B U_B^T = U_B^T U_B = I$$

$$\|A - BXC\|_F^2 = \left\| A - U_B \begin{pmatrix} \Sigma_B & 0 \\ 0 & 0 \end{pmatrix} V_B^T X U_C \begin{pmatrix} \Sigma_C & 0 \\ 0 & 0 \end{pmatrix} V_C^T \right\|_F^2$$

$$= \left\| \underbrace{U_B^T A V_C}_{\tilde{A}_{m\times n}} - \begin{pmatrix} \Sigma_B & 0 \\ 0 & 0 \end{pmatrix} \underbrace{V_B^T X U_C}_{\tilde{X}_{p\times q}} \begin{pmatrix} \Sigma_C & 0 \\ 0 & 0 \end{pmatrix} \right\|_F^2$$ 正交变换 F-norm 不变

$$= \|\underset{s\times t}{\tilde{A}_{11}} - \Sigma_B \underset{s\times t}{\tilde{X}_{11}} \Sigma_C\|_F^2 + \mathrm{Const}$$

$\min \|A - BXC\|$ s.t. $\mathrm{rank}(X) \le r$

$\min \|\tilde{A}_{11} - \Sigma_B \tilde{X}_{11} \Sigma_C\|$ s.t. $\mathrm{rank}(X) \le r$

$$\tilde{X}_{11}^{\#} = \Sigma_B^{-1} \cdot \mathrm{trunc_SVD}_r(\tilde{A}_{11}) \cdot \Sigma_C^{-1}$$

then $$X^{\#} = V_B \begin{bmatrix} \Sigma_B^{-1} \cdot \mathrm{trunc_SVD}_r([U_B^T A V_C]_{s,t}) \cdot \Sigma_C^{-1} & 0 \\ 0 & 0 \end{bmatrix} U_C^T$$

$\min \|y - Ax\|$ s.t. $\|x\|_0 \le K$

$$\cancel{x^{\#} = \begin{bmatrix} V\Sigma^{-1} H_K([U^H y]_t) \\ 0_{(n-t)\times 1} \end{bmatrix}}$$

正交变换不变秩，
但变 L_0-norm

$\min \|A - BXC\|$ s.t. $\|X\| \le \rho$

$$L = \|\tilde{A}_{11} - \Sigma_B \tilde{X}_{11} \Sigma_C\|^2 + \lambda(\|\tilde{X}_{11}\|^2 - \rho^2)$$

$$\nabla L = \Sigma_B(\Sigma_B \tilde{X}_{11} \Sigma_C - \tilde{A}_{11})\Sigma_C + \lambda \tilde{X}_{11} = 0 \qquad (2.2)$$

$$\tilde{x}_{ij}^{\#} = \frac{\tilde{a}_{ij}}{\sigma_{ij} + \lambda \sigma_{ij}^{-1}} \qquad \tilde{a}_{ij} = [\tilde{A}_{11}]_{ij}$$

$$\sigma_{ij} = \sigma_i(B)\,\sigma_j(C)$$

然后代 λ $\|\tilde{X}_{11}\|^2 = \rho^2$

数值解 λ，无闭式解

$\min \|y - Ax\|$ s.t. $\|x\| \le \rho$

$x^{\#} = (A^H A + \lambda I)^{-1} A^H y$ 由 ρ 求 λ

$\min \|A - BXC\|$ s.t. $X \in S$ closed convex (4.1)

用 AP 的思想

$$Y^{k+1} = \mathrm{Proj}_S(X^k - Z^k)$$

$$X^{k+1} = \arg\min \|A - BXC\|$$

s.t. $\|X - Y^{k+1}\| \le \rho$

用 Dykstra 的思想，变量 z 叫 Dykstra Correlation

(4.2) $Y^{k+1} = \mathrm{Proj}_S(X^k - Z^k)$

(4.3) $X^{k+1} = \arg\min \| A - BXC \|$ $\Longrightarrow$ $\min \| A' - BX'C \| + \lambda \| X' \|$

s.t. $\| X - Z^k - Y^{k+1} \| \le \rho$ $\quad W = Z^k - Y^{k+1}, \quad X' = X - W$

(4.4) $Z^{k+1} = Z^k - X^{k+1} + Y^{k+1}$ $\quad A' = A - BWC$

(4.1) 必有最优解 X^* 除了 PSD 约束，因为 PSD 不 close

Prop 4.1 前面即凸问题有解

收敛性证明

(4.1) $\longrightarrow \langle \nabla f^*, Y - X^* \rangle, \ \forall Y \in S$

(4.2) $\longrightarrow \langle \nabla f^k, Y - Y^{k+1} \rangle, \ \forall Y \in S$

(4.3) $\longrightarrow \nabla L^k = 0$

令 $\varepsilon^k = \| X^k - X^* \|^2 + \| Z^k - Z^* \|^2$

只需证 $\varepsilon^k - \varepsilon^{k+1} \ge$ 某个正数，故收敛

$$\| X^k - X^{k+1} \|^2 + \| Z^k - Z^{k+1} \|^2$$
$$+ \frac{2}{\lambda} \sigma_{\min}^2(B) \sigma_{\min}^2(C) \left(\| X^{k+1} - X^* \|^2 + \| X^{k+1} - X^k \|^2 \right)$$

线性收敛率

$$\varepsilon^k / \varepsilon^{k+1} \ge 1 + \delta \qquad \delta = 2 \left[\frac{\lambda}{\sigma_{\min}^2(B) \sigma_{\min}^2(C)} + \frac{\sigma_{\max}^2(B) \sigma_{\max}^2(C)}{\lambda} \right]^{-1}$$

$\lambda = \sigma_{\max}(B) \sigma_{\min}(B) \sigma_{\max}(C) \sigma_{\min}(C)$ 时最快

$$\min \| A - BXC \| \quad \text{s.t. } X \in S_1 \cap S_2$$

$$X^{k+1} = \operatorname{argmin} \| A - BXC \|^2 + \lambda \| X - W^k + Z_1^k \|^2$$

$$Y^{k+1} = \mathrm{Proj}_{S_1}(W^k - Z_2^k)$$

$$W^{k+1} = \mathrm{Proj}_{S_2}\left\{ \frac{1}{2}(X^{k+1} + Y^{k+1} + Z_1^k + Z_2^k) \right\}$$

$$Z_1^{k+1} = Z_1^k + X^{k+1} - W^{k+1}$$

$$Z_2^{k+1} = Z_2^k + Y^{k+1} - W^{k+1}$$